

Curating for Capability: A Low-CPU Framework for Building Specialized LLM Datasets

Defining and Acquiring High-Quality Curated Sources by Domain

The foundational principle of this research is the exclusive use of high-quality, curated sources to build specialized datasets for Large Language Model (LLM) training . This approach prioritizes data integrity, relevance, and reliability over sheer volume, which is particularly critical when operating within the strict computational limits of a low CPU speed environment . The definition of "high quality" varies by domain but consistently involves verifiable provenance, adherence to licensing agreements, and rigorous filtering to remove noise and bias [11](#) [19](#) . For general knowledge, a high-quality dataset often comprises a scaled-up version of heavily filtered publicly available web data, supplemented with licensed books and synthetic data [19](#) [147](#) . Microsoft's Phi-3 models, designed for resource-constrained settings, exemplify this strategy by being trained on a dataset composed of such filtered content [130](#)[164](#) . The Common Corpus is another example of a large-scale collection built on ethical data principles for pre-training [32](#) . However, even reputable sources like Wikipedia require careful handling; while its community-maintained reliability is a strength, it can also contain biases that compromise fairness if not addressed [165](#)[169](#) .

For conversational dialogue, quality is intrinsically linked to human-centricity, diversity, and safety. The OpenAssistant Conversation Dataset (OASST1) serves as a prime example, offering a human-generated, human-annotated corpus designed to democratize alignment research [116](#)[157](#) . More comprehensive resources like DialogStudio unify over 80 diverse dialogue datasets into a consistent format, saving significant curation effort [140](#) [141](#) . In sensitive contexts, such as mental health support, frameworks like Ethical Dialogue Modeling (EDM) are developed to produce safety-aware conversational agents [139](#) . Furthermore, emerging benchmarks emphasize best practices for consent, privacy, and compensation to ensure ethical AI development [120](#) . The challenge lies in balancing

semantic richness with these safety requirements, a task that necessitates sophisticated curation strategies 118139.

In the domain of code generation, high-quality data is characterized by structural correctness, functional relevance, and contextual clarity. Open-source repositories on platforms like GitHub represent a rich source of training material, with datasets such as the GitHub AI Repository Recommendation Dataset providing structured information about millions of projects 85 . Advanced curation techniques leverage LLMs themselves to score and filter code data, demonstrating a self-improving loop where models help refine their own training inputs 191. The presence of comments is a critical feature for enhancing model performance, as they provide functional context and can be used to augment existing datasets 160205. However, the use of open-source code introduces legal and ethical considerations, particularly concerning licenses like the GPL, which must be carefully navigated during data selection 45 . Benchmarks such as USENIX Security CCS 2024–2026 highlight the demand for specialized, high-quality data tailored to specific coding tasks 37 .

While less explicitly detailed than other domains, technical documentation requires sourcing from official APIs, developer guides, and technical articles. The primary challenge is extracting semantically meaningful chunks of text from these sources. Analogous to AST-based chunking for code 156 , parsing structured formats like Markdown or reStructuredText could yield more coherent training samples. The growing need for reliable evaluation of LLMs in technical domains underscores the importance of developing high-quality documentation datasets 14 . Across all domains, the key is to move beyond indiscriminate web scraping and target established, reputable repositories and communities to establish a high baseline of quality from the outset.

Domain	Key Characteristics of High-Quality Sources	Example Datasets/Repositories
General Knowledge	Heavily filtered public web data, licensed books, synthetic data; emphasis on reliability and broad coverage.	phi-2/phi-3 datasets 19 164 , Common Corpus 32 , filtered Wikipedia 169
Conversational Dialogue	Human-generated, human-annotated, diverse, and safe; may require ethical guidelines and formatting consistency.	OASST1 116 , DialogStudio 140 , FHIBE 120
Code Generation	Structurally correct, functionally relevant, with associated comments; requires attention to licensing and license compatibility.	Seed-Coder 191 , GitHub AI Repositories 85 , Mify-Coder benchmarks 79
Technical Documentation	Semantically coherent, sourced from official documentation and APIs; challenges in chunking and structuring.	Not explicitly listed, inferred from developer needs 14

Optimizing the Preprocessing Pipeline for Low-Resource Environments

In a low CPU speed environment, the data preprocessing pipeline is not merely a preparatory step but a critical subsystem whose efficiency directly determines the feasibility of the entire training project. The goal is to transform raw, curated text into a format ready for the model while minimizing computational overhead at every stage. A foundational optimization is the choice of data format. While plain text or JSONL are common, columnar formats like Apache Parquet offer superior storage efficiency due to higher compression ratios [108](#). More importantly for in-memory operations, Apache Arrow is engineered for low-overhead data interchange, defining a standardized memory layout that allows different systems to access data without serialization/deserialization costs [71](#) [72](#). The integration of a high-performance Rust implementation of a Parquet reader within the Arrow ecosystem further accelerates I/O, making this combination ideal for feeding data to a CPU-bound training loop and reducing latency [124125227](#).

A second major optimization addresses memory management. Traditional data loading often requires an entire dataset to be loaded into RAM before training can begin, a prohibitive requirement for large corpora [92](#). The solution is data streaming, a paradigm enabled by libraries like Hugging Face `datasets` through its `dataset streaming` feature. This allows the training loop to iterate over a dataset without downloading or storing the entire collection locally, dramatically reducing initial setup time and peak memory usage [146159](#). This is complemented by lazy loading patterns in custom scripts, where data items are generated and processed one at a time using constructs like Python's `yield`, preventing the creation of large intermediate lists in memory [92](#). Advanced implementations leverage memory-mapped files, where a dataset on disk is accessed as if it were in RAM, with the operating system's virtual memory manager loading only necessary pages into physical memory [218219](#). These techniques ensure the CPU is continuously supplied with data, preventing it from stalling while waiting for I/O operations to complete.

Tokenization, the process of converting text into numerical IDs, is a significant computational bottleneck in the end-to-end LLM pipeline [104207](#). Several strategies can mitigate this cost. Research suggests that developing a tokenizer in tandem with the model can lead to more efficient representations [46](#). For the code generation domain specifically, compiler-based preliminary tokenization has been shown to reduce input token counts by up to 33.7% compared to standard methods, offering a substantial reduction in downstream processing [192193](#). Adaptive tokenization schemes, such as

ChannelTok, which create flexible-length tokens, can also improve efficiency [1](#). Beyond tokenization, a robust preprocessing pipeline includes essential cleaning and normalization steps to reduce linguistic noise, which improves both model accuracy and computational efficiency [3](#) [25](#). Deduplication is another crucial step to ensure data diversity [25](#). Finally, advanced techniques like using LLMs for data augmentation—such as generating comments for code datasets to improve model understanding—are possible but introduce their own computational cost that must be carefully weighed against the expected improvement in model performance [132160](#). By combining optimized data formats, streaming architectures, and efficient tokenization methods, a highly performant preprocessing workflow can be constructed that respects the stringent constraints of a low-CPU system.

Selecting and Fine-Tuning Efficient Architectures for Constrained Hardware

The ultimate objective of dataset creation is to train an LLM, a process whose feasibility on low-CPU hardware is fundamentally dictated by the choice of model architecture and the training methodology. The evidence strongly indicates that Large Language Models (LLMs) are not the only viable option for this task; Small Language Models (SLMs) have emerged as a powerful and practical alternative for resource-constrained environments [53](#) [76](#). Models like Microsoft's Phi-3 series are explicitly designed to deliver high capability while maintaining cost-effectiveness, often outperforming models of the same size and even larger ones [103130](#). Other examples include TinyLlama, which has been successfully fine-tuned using CPU-friendly methods [131](#), and a range of SLMs evaluated on various coding benchmarks [50](#). These models strike a crucial balance between performance and resource consumption, making them the logical starting point for any project targeting low-CPU systems.

Once a suitable SLM is selected, the training process itself must be optimized. Standard full-parameter fine-tuning is computationally intensive and typically requires GPU acceleration. To overcome this, parameter-efficient fine-tuning (PEFT) methods are essential. Low-Rank Adaptation (LoRA) is repeatedly highlighted as a CPU-friendly technique that freezes the original model weights and injects trainable low-rank adapters, drastically reducing the number of parameters that need to be updated during training [131](#). This significantly lowers the memory and computational requirements, enabling fine-tuning on consumer-grade CPUs [131](#). Beyond PEFT, other architectural and optimization

strategies can further enhance efficiency. Quantization, which involves representing model weights with lower-precision data types (e.g., 4-bit integers), reduces memory pressure and can accelerate computation, especially on CPUs that support specific instruction sets [154](#)[182](#). Structured sparsity is another technique that prunes unnecessary connections in the model, leading to reduced memory usage for optimizer states and gradients, which in turn minimizes overall computational load [43](#). CPU offloading techniques, which dynamically shift parts of the computation between the CPU and GPU, can also be leveraged if some GPU is available, though the focus remains on minimizing reliance on it [166](#).

It is also important to consider the implications for inference, as the trained model will likely need to be run on the same constrained hardware. The CPU plays a central role in inference, handling not only model execution but also tasks like data transfer and tokenization, which can become significant bottlenecks [44](#) [207](#). Tokenization latency, in particular, can dominate the total cost of inference in high-throughput applications [207](#). Therefore, optimizations applied during the preprocessing phase, such as using efficient data formats (Parquet/Arrow) and optimized tokenization, directly translate to faster inference times. Furthermore, the final model, potentially already quantized, is better suited for fast execution on CPUs [182](#). The interplay between training and inference optimization is crucial; a well-curated dataset combined with a streamlined preprocessing pipeline ensures that the resulting model is not only capable but also deployable and responsive on the target low-resource platform.

Synthesis and Actionable Recommendations for Dataset Development

This research report outlines a strategic framework for developing high-quality, single-domain datasets for LLM training under the specific constraint of low CPU speed. The analysis confirms that success is contingent upon a holistic and optimized approach that treats the entire data lifecycle—from source acquisition to model deployment—as a single, interconnected system where efficiency at each stage compounds to enable the final goal. The core directive—to focus on one content type per dataset and exclusively use curated sources—is a sound strategy that establishes a high baseline of quality, thereby reducing the need for computationally expensive cleaning and filtering during preprocessing [19](#) [116](#)[191](#). This disciplined, focused approach aligns perfectly with the

limitations of resource-constrained hardware, as it avoids the overwhelming scale of mixed-domain, web-scraped datasets that are typical for training frontier models 11.

Based on the comprehensive analysis, a clear set of actionable recommendations emerges. First, data acquisition should prioritize reputable, pre-curated repositories and academic publications to ensure quality from the outset. For general knowledge, this means leveraging filtered web data and licensed texts 19; for dialogue, using unified datasets like DialogStudio 140; for code, utilizing repositories like GitHub and applying LLM-assisted filtering 85 191; and for technical documentation, focusing on structured API and guide sources 14. Second, the entire data handling infrastructure must be built around modern, efficient technologies. Adopting Apache Parquet for storage and Apache Arrow for in-memory interchange provides a powerful combination of storage savings and low-latency data loading, directly addressing the CPU bottleneck 72 108. Third, the preprocessing pipeline must be designed to operate within memory constraints. Implementing data streaming via tools like Hugging Face `datasets` is non-negotiable for managing large corpora, and lazy-loading patterns should be incorporated wherever possible to prevent memory overflow 92 146.

Finally, the project should commence with a small-scale proof-of-concept using a state-of-the-art Small Language Model (SLM) such as Phi-3-mini or TinyLlama, combined with a parameter-efficient fine-tuning method like LoRA 131164. This approach minimizes the computational burden of the training phase itself. The entire workflow must be tailored to the specific domain, with techniques like AST-based chunking for code or compiler-assisted tokenization being explored to maximize efficiency 156192. Before committing significant resources, it is imperative to benchmark the end-to-end pipeline—including data loading, preprocessing, and a single training step—on the target low-CPU hardware. This will establish a realistic performance baseline and identify any unforeseen bottlenecks. By adhering to this systematic, evidence-based strategy, it is entirely feasible to construct specialized, high-quality datasets that can effectively train capable LLMs within the demanding confines of a low-CPU environment.

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